UNIVERSITY OF WOLLONGONG IN DUBAI

UNDERGRADUATE PROGRAM SUBJECT OUTLINE

SCHOOL OF COMPUTER SCIENCES

http://my.uowdubai.ac.ae

SUBJECT NAME: NETWORK SECURITY

Course code:	CSCI368	Section:	1
Credit Points:	6	Year	2025
Session	Autumn	Duration:	2 hours lecture, 1 hour tutorial, 2 hours lab
Pre-requisite(s)	CSIT127, CSIT121, and 12cp 200 level CSCI/CSIT	Co-requisite(s)	
Mode of Delivery:	F2F Lectures/PPT/ICT		
Final Exam Passing Requirement:	40%		

LECTURE INFORMATION

Day:	Wednesday	Friday
Time:	14:30 – 16:30	16:30 – 18:30
Location:	3.44-Classroom B	5.11-Classroom B

Educator's Name:	Dr. Mouhannad ALATTAR
Building & Office No:	3.28
E-mail Address:	MouhannadAlattar@uowdubai.ac.ae
Consultation:	Wednesday: 12:30 – 14:30 Friday 14:30 – 16:30 Or you can send an email to request online meeting. https://uow.webex.com/meet/malattar

1 SUBJECT DESCRIPTION

This subject provides a survey of network security technologies and explores them in practice. This includes but is not limited to, network-based threats, security failure in cryptographic and network protocols, authentication servers, certificates and public-key infrastructures, security provisions in communication protocols and standards based on cryptographic methods, and electronic mail security.

2 CONTRIBUTION TO PROGRAM LEARNING OUTCOMES (PLO)

No.	Program Learning Outcomes
PLO1	Apply the core fundamentals of cyber security to real world situations
PLO2	Assess cyber security problems and vulnerabilities in software, computers, and networks and select appropriate security measures.

Upon successful completion of this subject, a student should be able to:		PLO 1	PLO 2
LO1	Describe and assess network vulnerabilities and network-based attacks		✓
LO2	Design and analyse security protocols, in particular those for key establishment and authentication, based on cryptographic methods and tools	✓	✓
LO3	Apply and use appropriate security standards and network security tools to enhance security of a distributed system	✓	✓
LO4	Evaluate, compare and recommend network security applications and configurations	✓	✓

3 SUBJECT SCHEDULE

Week	Lecture Topic(s)	Learning Outcomes	Session Type	Delivery format	Related supporting materials	Assessment Formative (F) Summative (S)
1	Introduction to Network protocols, network vulnerabilities, countermeasures	1, 4	L	Face to Face (F2F)	Slides, Quizzes, Class discussions	F
2	Key Agreement, Authentication (Kerberos…)	1, 2, 3	L	Face to Face (F2F)	Slides, Quizzes, Class discussions, Project work	F
3	Security at the application layer: Email Security, PGP, S/MIME	1, 2, 3, 4	L	Face to Face (F2F)	Slides, Quizzes, Class discussions, Project work	F
4	Transport Layer Security and SSL/TLS	1, 2, 3, 4	L	Face to Face (F2F)	Slides, Quizzes, Class discussions, Project work	F
5	Internet Layer Security and IPSec	1, 2, 3, 4	L	Face to Face (F2F)	Slides, Quizzes, Class discussions, Project work	F
6	Firewalls, VPN	1, 2, 3, 4	L	Face to Face (F2F)	Slides, Quizzes, Class discussions, Project work	F
Mid Semester Examination						

Week	Lecture Topic(s)	Learning Outcomes	Session Type	Delivery format	Related supporting materials	Assessment Formative (F) Summative (S)
7	Intrusion Detection Systems, Intrusion Prevention Systems	1, 2, 4	L	Face to Face (F2F)	Slides, Quizzes, Class discussions, Project work	F
8	Link-Layer and WiFi Security	1, 2, 3, 4	L	Face to Face (F2F)	Slides, Quizzes, Class discussions, Project work	F
9	Mobile Communication Security	1, 2, 3, 4	L	Face to Face (F2F)	Slides, Quizzes, Class discussions, Project work	F
10	E-commerce security Ad-hoc Network Security Cloud Security	1, 3, 4	L	Face to Face (F2F)	Slides, Quizzes, Class discussions, Project work	F
11	Review					

4 SUPPORTING MATERIALS

Books, Articles, Videos, Podcasts, etc. will be available on our Learning Management System (LMS)

4.1 REQUIRED TEXT

- W. Stallings, Cryptography and Network Security, 7th Edition, Pearson, 2016.
- William Stallings, Network Security Essentials, 6th edition, Pearson, 2016.

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4.2 RECOMMENDED READING

- IBM Cost of a Data Breach Report 2025
- C. Kaufman, R. Perlman, and M. Speciner, Network Security: PRIVATE communication in a PUBLIC world,2nd edition, Prentice Hall, 2002
- C. Boyd, A. Mathuria, D. Stebila, Protocols for Authentication and Key Establishment, 2nd Edition, Springer, 2020
- P. Campbell, B. Calvert, 2003. Security+ Guide to Network Security Fundamentals, Thomson
- M. Ciampa, 2008, Security+ Guide to Network Security Fundamentals. Cengage Learning
- W. Stallings, 2010, 5th edition, Cryptography and Network Security: Principles and practices, Pearson

4.3 LECTURE NOTES

All teaching material can be downloaded from the Moodle site for this subject.

4.4 ACCESS TO SUPPORTING MATERIALS

The University uses MOODLE as a Learning Management System (LMS) to support all coursework subjects. The subject site and supporting materials can be accessed via: https://moodle.uowplatform.edu.au and via UOWD Library.

5 ASSESSMENT

5.1 ASSESSMENT OF LEARNING OUTCOMES

Learning Outcome	Assessment 1 Labs 20%	Assessment 2 Tutorial Project 20%	Assessment 3 Midterm 20%	Assessment 4 Final Exam 40%
LO1	✓	✓	✓	✓
LO2	✓	✓	✓	✓
LO3	✓	✓	✓	✓
LO4	✓	✓	✓	✓

5.2 USING GENERATIVE ARTIFICIAL INTELLIGENCE (GENAI)

GenAI technology (such as ChatGPT or Microsoft Co-pilot) is reshaping the University experience worldwide. UOW is committed to embracing GenAI as a tool to enhance learning experiences and develop vital work-readiness skills. However, misuse or use of GenAI in assessments where prohibited constitutes academic misconduct (as specified by University Policy).

It is important that students check if GenAI is permitted for each assessment task and how it is to be used and acknowledged. Please read the student guidance (Acknowledging the permitted use of GenAI in assessments, Using Generative AI tools well and Access your UOW Microsoft Copilot account) available on how to use GenAI ethically and critically, equally recognising its capabilities and limitations. For example:

1. **Generative AI is not a substitute for decision-making**: GenAI should complement, not replace, your critical thinking and decision-making skills.
2. **Output quality depends on prompts**: The quality of GenAI outputs is influenced by prompting. Poorly constructed or unclear prompts may generate outputs that are incorrect.
3. **Fact verification is essential**: GenAI outputs can be fabricated, presenting inaccurate information or contain harmful bias. Verify all GenAI outputs against reliable sources.
4. **Protect data and copyright**: Many GenAI technologies collect information in ways that breach privacy and data protection provisions, particularly where the source material is confidential or subject to copyright. Please check the Terms and Conditions of GenAI technologies and if unsure, contact UOW Copyright Guidance. Learn more about how to access UOW secured GenAI tools here.
5. **Transparency in use**: Where required, you must acknowledge GenAI use, including providing prompt histories and detailing how GenAI was utilised.
6. **Thoughtful and appropriate application**: Be mindful of when and how to use GenAI tools. Assess its appropriateness for each use, and refrain from use when not suitable.

5.3 ASSESSMENT TASKS

Assessment Task:	Labs
Type:	Individual and/or Group

Description:	The lab sessions are hands-on practical exercises where students work under the supervision of an instructor to apply the concepts learned in lectures and tutorials. The lab environment encourages students to practice key network security skills, gain familiarity with various network tools, and understand how theoretical concepts are applied in practice.
Learning Outcome Measured:	All
Total Marks	20
Weighting:	20%
Due Date:	Throughout the trimester, starting week 2
TurnItIn submission required by:	Student

OUTLINE AND REQUIREMENTS

There will be weekly labs in this subject, many of them with submissions. If there is a submission, the lab sheet will include the submission date, which is usually before the end of the lab session or within a few days.

Your completed attempts must be submitted electronically as will be instructed in each lab sheet. Each submission MUST be fully commented and be ready to be reproduced.

- All labs are to be completed independently. Plagiarism may result in a zero mark being recorded for that submission and the case brought to the University Academic Integrity Committee.
- Any changes to submissions' dates will be posted on the website of the subject and/or communicated through email. It is each student's responsibility to regularly check the subject's website and their SOLS emails.
- Late submissions will incur 5% penalty/day, including weekends. Work more than 4 days late will be awarded a mark of zero.

Each submission should include a "Readme.txt" file that indicates any special information required to understand the submission as well as guidelines on how to reproduce the solution and asses the submitted material. Collaboration is encouraged for learning purposes, but each student must submit their own work unless a group submission is explicitly allowed.

MARKING CRITERIA

- Submissions that do not fully work/address the problem will not receive more than 40% of the marks.
- Marks are awarded based on closeness to final answer and correctness of the submission.
- Simplicity of the solution as well as its cost will be awarded marks.

Assessment Task:	Tutorial Project
Type:	Group
Description:	In this group project, students will design and implement a secure network for a hypothetical company using network simulation tools like Packet Tracer or similar platforms. The company's network will include a mix of on-premises sites, cloud resources, partner systems, etc. mimicking real-world business scenarios. Intermediate submissions or quizzes could be requested by the instructors
Learning Outcome Measured:	All

Total Marks	20
Weighting:	20%
Due Date:	Throughout the trimester, starting week 2
TurnItIn submission required by:	Student

OUTLINE AND REQUIREMENTS

Each group will design and implement the network infrastructure for a hypothetical company. As the course progresses, you will be required to **update your network design** and **security controls**, reflecting new concepts such as key agreements, email security, transport layer security (SSL/TLS), and firewalls. These updates should show how each new lecture topic influences your evolving network.

Intermediate submissions, exercises and quizzes can be requested by the instructors.

Each group must provide a **final report** that consolidates all weekly updates into a comprehensive view of the network.

- All projects are to be completed in group. Plagiarism may result in a zero mark being recorded for that project and the case brought to the Academic Integrity Committee.
- Any changes to submissions' due dates will be posted on the website of the subject and/or communicated through email. It is each student's responsibility to regularly check the subject's website and their SOLS emails.
- Late submissions will incur 5% penalty/day, including weekends. Work more than 4 days late will be awarded a mark of zero.
- Each submission should include full documentation of new progresses, and a "readme" file that indicates any special information required to understand the submission as well as guidelines on how to reproduce the solution and asses the submitted material.

MARKING CRITERIA

- Submissions that do not fully meet the project requirements or address the stated network and security challenges discussed in lectures and tutorials will not receive more than 40% of the marks.
- Marks will be awarded based on:
    - simplicity and how closely the submission aligns with the expected solution in terms of correctness, configuration accuracy, and how well the network addresses the evolving challenges discussed during lectures and tutorials.
    - quality of the report that provides network diagram and explanations of security decisions.
    - quality of final presentation (if requested by the instructor) summarizing the design, challenges faced, and how security was implemented.

Assessment Task:	Mid-Session Examination (Midterm)
Type:	Individual
Description:	A midterm to test the learning outcomes in this subject
Learning Outcome Measured:	All
Total Marks	50
Weighting:	20%
Due Date:	To be held during the official midterm examination period. Please refer to the Exam Timetable available on the Student Online Resources website (http://my.uowdubai.ac.ae) closer to the exam period.
TurnItIn submission required by:	

OUTLINE AND REQUIREMENTS

There will be a midterm to test the learning outcomes in this subject. The midterm will consist of short/elaborative answers that will run starting week 1 all the way till the end of the Week 6. The material covered to-date in lectures, tutorials, and labs. The goal of this is to assess students' learning at various points in the semester. Students' marks in this assessment are usually a good indication of the performance of students in the final exam and overall assessments. The tests are an individual assessment and take place under exam conditions.

MARKING CRITERIA

- Marks distribution of each question is going to be included with the exam sheet.
- Answers will be checked against appropriateness of answer.

Assessment Task:	Final Examination
Type:	Individual
Description:	Final examination
Learning Outcome Measured:	All
Total Marks	50
Weighting:	40%
Due Date:	To be held during the official examination period. Please refer to the Exam Timetable available on the Student Online Resources website (http://my.uowdubai.ac.ae) closer to the exam period.
TurnItIn submission required by:	

OUTLINE AND REQUIREMENTS

This assessment will test learning outcomes of this subject from week 7 to week 11. The final is a comprehensive exam that covers material from lectures, tutorials, assignments, project, and labs.

MARKING CRITERIA

- Marks distribution of each question is going to be included with the exam sheet.
- Answers will be checked against appropriateness of answer.

SUBMISSIONS:

Please note that late submissions will incur a penalty of 5% per day, including weekends. Work more than 4 days late will be awarded a mark of zero.

5.4 GRADES AWARDED

The approved grades of performance and associated ranges of marks for undergraduate subjects are:

High Distinction (HD)	85 – 100%
Distinction (D)	75 – 84%
Credit (C)	65 – 74%
Pass (P)	50 – 64%
Pass Supplementary (PS)	50%
Fail (F)	0 – 49% (and not meeting the attendance requirements)
Technical Fail (TF)	Not meeting the final exam passing requirements – see the Assessment Policy PP-REG-DB-2.1

5.5 SATISFACTORY COMPLETION REQUIREMENTS

In order to be considered for a grade of Pass (P) or better in this subject, students must achieve the minimum required mark in the Final Examination (see page 1 for required score); students who obtain a composite mark greater than or equal to 50% but do not satisfy the Final Examination minimum pass requirements in the final examination will be awarded a "Technical Fail" grade.

Students must 'reasonably' complete all assessment tasks (including the required score for the Final Examination,) and submit these as specified in the subject outline. 'Reasonable' completion of an assessment task will be determined based on the instructions given to the student including: word length, demonstration of research and analysis where required, adherence to the Plagiarism Policy guidelines, and completion of each section/component of the assessment. Failure to submit all assessment tasks may result in a Fail grade awarded for the subject.

6 RELEVANT POLICIES AND DOCUMENTS

All students must read and be familiar with the following UOWD policies and documents, which are available on the Student Online Resources (my.uowdubai.ac.ae) website by following the Policies link:

- Academic Grievance Policy
- Academic Integrity Policy
- Campus Access and Order Rules
- Code of Conduct – Library Users
- Code of Practice – Students
- Copyright Policy
- Intellectual Property Policy
- Library Regulations
- Minimum Rate of Progress
- Music, Video and Software Piracy
- Non-Discriminatory Language and Practice & Presentation Policy and Guidelines
- Special Consideration Policy & Procedure
- Student Attendance Policy
- Student Conduct Rules
- Rules for use of UOWD ITTS Facilities
- Teaching and Assessment: Code of Practice – Teaching
- Teaching and Assessment: Assessment and Feedback Policy
- Teaching and Assessment: Subject Delivery Policy

7 SSP & STUDIOSITY

SSP (Student Support Program) is a program committed to assisting students in developing their academic skills and getting the most out of their studies. As part of their services, SSP provides Peer Tutoring Program and Academic Workshops (https://my.uowdubai.ac.ae/ssd/index.php).

Studiosity is an online study tool that students can access 24 hours, 7 days a week! Students can receive feedback on submitted writing in less than 24 hours and receive one-to-one, personal help in real time with a subject specialist. The service can be accessed through the subject's Moodle site.

For further information, please contact:
SSP Coordinator
Room 020, Block 5.
ssp@uowdubai.ac.ae

Phone Number: +971 4 278 1756

8 ACADEMIC INTEGRITY

Plagiarism and cheating are serious offences that can lead to expulsion from the university. Students must be familiar with the *Academic Integrity* policy which outlines the procedure that will be followed in case of academic misconduct including cheating and plagiarism. Please refer to *How to Avoid Plagiarism* available on the Student Online Resources website (http://my.uowdubai.ac.ae).

8.1 TURNITIN

Students are required to submit all written assignments in soft copy through the TurnItIn system which is available online at www.turnitin.com. Every student must have a TurnItIn account. Failure to submit an assignment through TurnItIn will result in marks for that assignment being withheld. **Students do NOT need to hand in a printed copy of the TurnItIn Originality Report.** More information about TurnItIn (including how to create an account and add a class) will be provided in the first lecture. Students can download Frequently Asked Questions (FAQs) about TurnItIn from the SSP section of UOWD website (https://www.uowdubai.ac.ae/academic-resources/student-support-programs).

TurnItIn information required to add this subject:

Class ID:	Moodle Link
Password:	Moodle Link

8.2 REFERENCE & IN-TEXT CITATION

For information about referencing and in-text citation please review the *Academic Writing Presentation* available on the Student Online Resources website (http://my.uowdubai.ac.ae).

8.3 UOWD RULES & POLICIES

For information about UOWD Rules and Policies, please go to the Student Online Resources website (http://my.uowdubai.ac.ae) and click on the POLICIES link.

9 ATTENDANCE REQUIREMENTS

Attendance in this subject is compulsory. Failure to attend all tutorials and computer labs as per the Student Attendance Policy may result in a FAIL grade. Students are strongly encouraged to become familiar with this policy (which can be found on the Online Resources website at my.uowdubai.ac.ae).

10 TUTORIAL/COMPUTER LAB ENROLMENTS

All students must sign up for one tutorial and/or computer lab in Week 1. Admission to a tutorial/computer lab will not be possible unless the student's name is on the Attendance List for that class. No changes will be allowed once a student has enrolled in a tutorial/computer lab.

11 SUPPLEMENTARY ASSESSMENTS

A supplementary assessment may be offered to students whose performance in this subject is close (45-49 in the final examination and 48-49 in the composite score) to that required to pass the subject, and are otherwise identified as meriting an offer of a supplementary assessment. The precise form of a supplementary assessment will be determined at the time the offer of a supplementary is made.

12 SUSTAINABILITY

UOWD encourages all students to act in a sustainable manner when planning and submitting assessments. If possible, students should not use plastic items, such as folders, covers, and bindings, and other synthetic materials, for presentations, workshops, and other activities. Students are also encouraged to avoid unnecessary printing; and if printing is required, please consider printing double-sided and only printing essential illustrations avoiding blocks of any colour as the use of ink is harmful to the environment. Always behave in a sustainable way.